

Project Report: Concealed Carry Laws and Crime

Do "Shall-Issue" Laws Reduce Violent Crime? Evidence from U.S. Panel Data (1977–1999)

1. Objective

This project evaluates whether "shall-issue" concealed carry laws affect violent crime rates using U.S. state-level panel data from 1977 to 1999 (1,173 observations). The focus is on understanding how **model choice affects conclusions** and why naive estimates can be misleading.

2. Data

2.1 Data Description

The dataset is a balanced panel covering 51 states (including Washington D.C.) from 1977 to 1999, totaling 1,173 observations (51 states $\times$ 23 years).

2.2 Descriptive Statistics

Variable	Description	Obs	Mean	Std Dev	Min	Max
year	Year	1,173	88.00	6.64	77	99
vio	Violent crime rate (per 100,000)	1,173	503.07	334.28	47.0	2,921.8
mur	Murder rate (per 100,000)	1,173	7.67	7.52	0.2	80.6
rob	Robbery rate (per 100,000)	1,173	161.82	170.51	6.4	1,635.1
shall	Shall-issue law indicator	1,173	0.24	0.43	0	1
incarc_rate	Incarceration rate	1,173	226.58	178.89	19.0	1,913.0
avginc	Real per capita income (\$1,000)	1,173	13.72	2.55	8.55	23.65
pop	State population (millions)	1,173	4.82	5.25	0.40	33.15
density	Population density (per 1,000/sq mi)	1,173	0.35	1.36	0.0007	11.10

3. Methodology

The analysis follows a step-by-step strategy to isolate causal effects:

Step	Model	Purpose
1	Simple OLS	Baseline estimate
2	OLS with Controls	Adjust for observable confounders
3	State Fixed Effects	Control for time-invariant state characteristics
4	Two-Way Fixed Effects	Control for state + time heterogeneity

All standard errors are clustered at the state level (51 clusters) to account for within-state correlation and serial correlation.

3.1 Econometric Specifications

Model 1: Simple OLS

$$\ln(\text{vio_st}) = \beta_0 + \beta_1 \cdot \text{shall_st} + \varepsilon_{\text{st}}$$

Model 2: OLS with Controls

$$\ln(\text{vio_st}) = \beta_0 + \beta_1 \cdot \text{shall_st} + \gamma \cdot X_{\text{st}} + \varepsilon_{\text{st}}$$

Model 3: State Fixed Effects

$$\ln(\text{vio_st}) = \beta_1 \cdot \text{shall_st} + \gamma \cdot X_{\text{st}} + \alpha_s + \varepsilon_{\text{st}}$$

Model 4: Two-Way Fixed Effects (State + Year)

$$\ln(\text{vio_st}) = \beta_1 \cdot \text{shall_st} + \gamma \cdot X_{\text{st}} + \alpha_s + \lambda_t + \varepsilon_{\text{st}}$$

4. Results

4.1 Main Results

Table 1: Effect of Shall-Issue Laws on Violent Crime

	(1) OLS	(2) OLS+Ctrl	(3) State FE	(4) Two-Way FE
Shall	-0.443*** (0.157)	-0.368*** (0.114)	-0.046 (0.042)	-0.028 (0.041)
Incarc_rate		0.002*** (0.001)	-0.000 (0.000)	0.000 (0.000)
Density		0.027 (0.041)	-0.172 (0.138)	-0.092 (0.124)
Avginc		0.001 (0.024)	-0.009 (0.013)	0.001 (0.016)
Pop		0.043*** (0.012)	0.012 (0.014)	-0.005 (0.015)
Observations	1,173	1,173	1,173	1,173
R² (within)	0.087	0.564	0.218	0.418
State FE	No	No	Yes	Yes
Year FE	No	No	No	Yes

Standard errors clustered at state level in parentheses

*** p<0.01, ** p<0.05, * p<0.1*

4.2 Key Finding

The effect becomes statistically insignificant when state fixed effects are added.

Specification	Coefficient	p-value	Interpretation
Simple OLS	-0.443	0.007	Large, significant
OLS with Controls	-0.368	0.002	Still large, significant
State Fixed Effects	-0.046	0.275	Small, not significant
Two-Way Fixed Effects	-0.028	0.495	Small, not significant

This progression demonstrates that naive OLS estimates are driven by omitted state-level heterogeneity rather than the causal effect of the law.

4.3 Interpretation

At first glance, the OLS results suggest a strong negative relationship between shall-issue laws and crime.

However, this result becomes statistically insignificant once fixed effects are introduced.

This was initially surprising, given the large magnitude of the OLS estimates.

After controlling for state-level heterogeneity, it becomes clear that:

- The initial results are driven by **omitted variable bias**
- States adopting the law are systematically different
- The observed relationship is not causal

In other words, the apparent effect reflects **cross-state differences**, not the impact of the policy itself.

4.4 Results by Crime Type

Table 2: Effects on Robbery and Murder (Two-Way FE)

Crime Type	Coefficient	p-value	95% CI	Interpretation
Robbery	0.027	0.609	[-0.078, 0.132]	+2.7%, not significant
Murder	-0.015	0.697	[-0.092, 0.062]	-1.5%, not significant

Neither robbery nor murder shows a statistically significant effect.

4.5 Model Selection: Hausman Test

Table 3: Hausman Test Results

Test	χ^2	df	p-value	Conclusion
Standard Hausman	41.63	29	0.0606	Weak evidence
Hausman (sigmamore)	80.19	8	<0.001	FE preferred

5. Interpretation

5.1 Why OLS Fails

The progression of estimates reveals why naive OLS is misleading:

- OLS shows large, significant effects
- Fixed effects remove significance

This indicates strong omitted variable bias driven by time-invariant state characteristics.

5.2 What This Means

The data does not provide robust evidence that shall-issue laws reduce crime.

The initial significant results are driven by:

- Cross-state differences
- Not by the causal effect of the law

6. Threats to Internal Validity

Threat	Description
Time-varying omitted variables	State-specific changes not captured by fixed effects
Endogenous timing	Policy adoption may respond to crime trends
Measurement error	Binary indicator may not reflect intensity
Spillover effects	Crime may shift geographically
Dynamic effects	No lag/anticipation modeling

7. Broader Implications

Domain	Example
Marketing	Campaign vs pre-existing growth
Product	Feature vs user selection
Policy	Treatment vs baseline differences

Key lesson: **Correlation is not causation**

8. Conclusion

This analysis demonstrates three key findings:

1. **Naive OLS estimates are misleading:** Simple regressions show large, significant effects (-44% to -31%)
2. **Fixed effects matter:** After controlling for state characteristics, the effect becomes small and statistically insignificant (-4.5% to -2.8%, $p > 0.27$)
3. **No robust evidence:** The data does not support the claim that shall-issue laws reduce violent crime, robbery, or murder

Final Conclusion:

The observed correlation between shall-issue laws and lower crime rates in simple regressions is driven by omitted state-level heterogeneity, not by the causal effect of the laws themselves.

9. Next Steps

- Event study
- Instrumental variables
- Heterogeneous effects
- Causal machine learning